

The empirical recurrence for OEIS A209811, proved by counting patterns

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A209811 counts arrays subject to a local condition *and* to the clause “new values 0..3 introduced in row major order”, which says the array is written in canonical form. That clause is not local, so a transfer matrix over the alphabet does not see it. It does not have to: what is being counted is not arrays but equality patterns, and the number of patterns is recovered from the numbers L_i of arrays over an i -letter alphabet by inverting a falling factorial. The result is the clean identity $4!a(n) = \sum_i \binom{4}{i} D_{4-i} L_i(n)$ with D_m the derangement numbers, and each L_i is an ordinary walk count. A second reduction — the chain is strongly lumpable over the patterns, because the condition is relabelling-invariant — brings the state space down from 4890 to $S = 342$. The entry’s empirical recurrence of order 91, contributed by R. H. Hardin, then follows from a finite exact computation.

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1 The sequence and the conjecture

OEIS A209811 is “Number of $(n+1) \times 6$ 0..3 arrays with every 2×2 subblock having two distinct values, and new values 0..3 introduced in row major order.”. It has offset 1 and begins

5193, 244208, 11912459, 617966001, 32449786403, 1723218327331, 91909548662623, 4917823780419687, ...

The entry carries the following comment:

Empirical: $a(n) = 107*a(n-1) - 1878*a(n-2) - 137789*a(n-3) + 4495053*a(n-4) + 60979281*a(n-5) - 3382378658*a(n-6) - 5378727169*a(n-7) + 1356204907007*a(n-8) - 5005172801248*a(n-9) - 338085898941018*a(n-10) + 2348610841442425*a(n-11) + 56536290959562784*a(n-12) - 541259839742806307*a(n-13) - 6608801096459681129*a(n-14) + 81395189596556559983*a(n-15) + 548394351778255652595*a(n-16) - 8770207024501159865529*a(n-17) - 31666973586868041812990*a(n-18) + 710376576458807557547216*a(n-19) + 1128924193484746151881180*a(n-20) - 44551314523662218100563425*a(n-21) - 7263266321979453932360270*a(n-22) + 2206376031045169589972758106*a(n-23) - 2026353407686910357730744032*a(n-24) - 874544724381776272260325) + 157141357413725898086880662843*a(n-26) + 2799017239399371977020592788297*a(n-27) - 7178294228784280740564781901801*a(n-28) - 72691433711711173410574983496660*a(n-29) + 239443617254126262209848548396745*a(n-30) + 1533257431497069166591149239829307*a(n-31) - 6221922488596066027751549217359670*a(n-32) - 26159175017183962402068162814758501*a(n-33) + 129627573400367716297066655136898331*a(n-34) + 356671191170640145323147937051920800*a(n-35) - 2198351761876285273302384497226420197*a(n-36) - 3777310666553059903322898621430359569*a(n-37) + 30602786767129664524288240851030652355*a(n-38) + 28834501346077162987374441766139145544*a(n-39) - 351203971753422209686115577891709785746*a(n-40) - 116701428260928979532142632668200005284*a(n-41) + 3327353660864163071142600026522029460808*a(n-42) - 560017644613994507027394206455325197980*a(n-43) - 25995896123168019862614781805551174366491*a(n-44) + 15575168550871264099563278351155746483043*a(n-45) + 166872533834247680226333166810473007305778*a(n-46) - 165583631919347715240453270733787800300967*a(n-47) - 874071453968014666788967372998585172431989*a(n-48) + 1206235128308924857940819522655801838918602*a(n-49) + 3692161451201415426488274381201249258574504*a(n-50) - 6665858797205359780634341310843509685136667$

51) $-12319864530826648929075899837324201902878036^*a(n-52) + 2890154560248637934842234942381086355682409$
 53) $+31170391469749070397533351890857301477542909^*a(n-54) - 995026744059640277800608471353947335153381$
 55) $-53857770398272002266481193998251821786322739^*a(n-56) + 272482776349046561779905216334225226036875$
 57) $+37540447794350619447986912229887791121752431^*a(n-58) - 590415367694103245739906836722138721711566$
 59) $+110686266599666467047553539311810235014535318^*a(n-60) + 1000112928372531572212817768326632301093$
 61) $-473363054517674986586028935552949302192543622^*a(n-62) - 129604922279405107310436966000245190537731$
 63) $+983069991779873415748977900612905249562346419^*a(n-64) + 1234590759437986624743595568834744319557$
 65) $-1367316646006905669194495899803555096031034679^*a(n-66) - 79032943041429220182666949684054738313241$
 67) $+1352458505939904189893228637165571481776553934^*a(n-68) + 242440486825297403697854903558758025775$
 69) $-955596064883055418717334290995333836848088796^*a(n-70) + 91237272508018656852555859600497039867416$
 71) $+470364980986922495221701297389250520645203036^*a(n-72) - 14941461156172283449700296790608348112521$
 73) $-151612194930909389895225913796358009798829200^*a(n-74) + 83459176699483481099731422805986592523856$
 75) $+27203058188450643926088113201093410799384736^*a(n-76) - 262665259413654955659850418944045103623353$
 77) $-835328187570826089554010524357221294549056^*a(n-78) + 4821949963807386895787459003961042725066880^*$
 79) $-679112903590477098800215366046658653702208^*a(n-80) - 478646169765785125170403007240661020667264^*a(n$
 81) $+134388350157539529383186764724725440221184^*a(n-82) + 19845613671413486485308605949798443768832^*a(n$
 83) $-10375428852018050542986066625716287592960^*a(n-84) + 118385596283169479151035951638780557312^*a(n-$
 85) $+342248518750406303246673137377100599296^*a(n-86) - 26440425803133463527797481814195814400^*a(n-$
 87) $-4234051645813953234637647944641339392^*a(n-88) + 454953491082351367104806724121657344^*a(n-$
 89) $+13225918069327774097709415191674880^*a(n-90) - 1738900086789955366482634541629440^*a(n-$
 91)

As of the “Last modified” line on the live entry (Oct 03 2025 15:30:42, revision 9) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) = 107a(n-1) - 1878a(n-2) - 137789a(n-3) + 4495053a(n-4) + 60979281a(n-5) - 3382378658a(n-6) - 5 \quad (1)$$

for all $n > 91$.

2 What the canonical-form clause counts

Fix the shape: 6 columns and $L = n + 1$ rows, so $L \cdot 6$ cells. Call a partition of the cells into nonempty classes a *pattern*; an array over an alphabet determines a pattern, two cells lying in the same class exactly when they carry the same value.

Every 2×2 subblock of the array occupies two consecutive rows and two consecutive columns; writing those two rows as r and s the subblock is

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} r_j & r_{j+1} \\ s_j & s_{j+1} \end{pmatrix},$$

and the entry’s requirement on it is

$$\#\{\alpha, \beta, \gamma, \delta\} \in \{2\}. \quad (2)$$

The condition is stated purely in terms of equalities between entries, so it is invariant under any relabelling of the values and is therefore a property of the pattern alone. Call a pattern *admissible* when it has the property.

The clause “new values 0..3 introduced in row major order” selects, from each class of arrays that differ only by a relabelling, exactly one representative: reading the cells in row major order, the first occurrence of the value v must precede the first occurrence of $v + 1$. Every pattern with at most $K = 4$ classes has exactly one such representative, and every array in canonical form is the representative of its own pattern. Hence, writing $N_j(n)$ for the number of admissible patterns with exactly j classes,

$$a(n) = \sum_{j=0}^4 N_j(n). \quad (3)$$

Lemma 1. *Let $L_i(n)$ be the number of arrays over an alphabet of i letters satisfying the local condition, with no canonical-form clause. Then*

$$L_i(n) = \sum_j N_j(n) i(i-1) \cdots (i-j+1), \quad 4! \sum_{j=0}^4 N_j(n) = \sum_{i=0}^4 \binom{4}{i} D_{4-i} L_i(n),$$

where $D_m = m! \sum_{t=0}^m (-1)^t / t!$ is the number of derangements of m letters.

Proof. An array over i letters satisfying the condition determines an admissible pattern together with an injection of its classes into the alphabet, and conversely; a pattern with j classes admits $i(i-1) \cdots (i-j+1)$ injections. That is the first identity. Writing $i(i-1) \cdots (i-j+1) = j! \binom{i}{j}$ it reads $L_i = \sum_j (j! N_j) \binom{i}{j}$, so by binomial inversion $j! N_j = \sum_i (-1)^{j-i} \binom{j}{i} L_i$. Summing over $j \leq 4$ and exchanging the order of summation,

$$\sum_{j=0}^4 N_j = \sum_{i=0}^4 L_i \sum_{j=i}^4 \frac{(-1)^{j-i}}{i! (j-i)!} = \sum_{i=0}^4 \frac{L_i}{i!} \sum_{t=0}^{4-i} \frac{(-1)^t}{t!},$$

and multiplying by $4!$ turns the coefficient of L_i into $\frac{4!}{i!} \sum_{t \leq 4-i} (-1)^t / t! = \binom{4}{i} (4-i)! \sum_{t \leq 4-i} (-1)^t / t! = \binom{4}{i} D_{4-i}$, an integer. $\square$

3 Each L_i is a walk count, and the chain lumps

Fix i and let $V_i = \{0, \dots, i-1\}^6$ be the possible rows. For $r, s \in V_i$ declare $r \rightarrow s$ an edge when placing s directly after r satisfies (2) at every column, and let M_i be the adjacency matrix. An array is exactly a sequence of rows, and its condition is exactly the conjunction of the edge conditions along that sequence, so

$$L_i(n) = \mathbf{s}_i^\top M_i^n \mathbf{1}.$$

Because the condition involves no row before the first, every row may start a walk, and $\mathbf{s}_i = \mathbf{1}$.

Lemma 2. *Let $\pi(r)$ be the equality pattern of the row r . For all r, r' with $\pi(r) = \pi(r')$ and every pattern Q ,*

$$\#\{s : r \rightarrow s, \pi(s) = Q\} = \#\{s : r' \rightarrow s, \pi(s) = Q\}.$$

Consequently the numbers $\widehat{M}_i[P][Q]$ obtained by taking any representative of P are well defined, and with $e_P = i(i-1) \cdots (i-|P|+1)$ the number of rows of pattern P ,

$$\mathbf{s}_i^\top M_i^m \mathbf{1} = \sum_P \varepsilon_P f_m(P), \quad f_0 \equiv 1, \quad f_{m+1}(P) = \sum_Q \widehat{M}_i[P][Q] f_m(Q),$$

where $\varepsilon_P = e_P$ if rows of pattern P may start a walk and 0 otherwise.

Proof. If $\pi(r) = \pi(r')$ then the map sending the value of r at a column to the value of r' at that column is a well-defined injection between the letters used, and extends to a permutation σ of the alphabet with $r' = \sigma(r)$. The edge condition is a statement about equalities among entries, so $r \rightarrow s$ if and only if $\sigma(r) \rightarrow \sigma(s)$; and $\pi(\sigma(s)) = \pi(s)$. Hence $s \mapsto \sigma(s)$ is a bijection between the two successor sets preserving patterns, which is the first claim. It says the chain is strongly lumpable over π , so $(M_i^m \mathbf{1})(r)$ depends on r only through $\pi(r)$; calling that value $f_m(\pi(r))$ gives the recursion, and summing over r in each class gives the displayed identity, the number of rows of pattern P being the number of injections of its $|P|$ classes into the alphabet. $\square$

Assembling the 4 lumped chains into one block-diagonal matrix N of size $S = 342$ and putting the weights of Lemma 1 into the start vector, $\mathbf{v} = (\binom{4}{i} D_{4-i} \varepsilon_P)_{i,P}$, gives

$$4! a(n) = \mathbf{v}^\top N^n \mathbf{1}. \quad (4)$$

The unlumped alphabet had 4890 rows in all; the lumped chain has $S = 342$ states.

4 The criterion, and the computation

Let $q(t) = t^{91} - \sum_i c_i t^{91-i}$ be the characteristic polynomial of (1).

Lemma 3. *For every n with $m = n + 1 - 1 \geq 91$,*

$$4! \left(a(n) - \sum_i c_i a(n-i) \right) = \mathbf{v}^\top N^{m-91} q(N) \mathbf{1}.$$

Hence (1) holds from that point on if and only if $\mathbf{v}^\top N^j q(N) \mathbf{1} = 0$ for every $j \geq 0$, and it is enough to check $j = 0, 1, \dots, S-1$.

Proof. Substitute (4) into the left side and factor N^{m-91} out of $N^m - \sum_i c_i N^{m-i}$. The scalar $4!$ is nonzero, so it does not affect whether the left side vanishes. For the last clause put $w = q(N) \mathbf{1}$: by Cayley–Hamilton N^S is an integer combination of $I, N, \dots, N^{S-1}$, so each $\mathbf{v}^\top N^j w$ with $j \geq S$ is the corresponding combination of earlier values. $\square$

Theorem 4. *The recurrence (1) holds for every $n > 91$, which is the statement of Section 1.*

Proof. The lumped transition counts were obtained by enumerating, for one representative row of each pattern, all admissible successors and sorting them by pattern — the successors being generated column by column, since the edge condition is a conjunction of one constraint per column. The vector $w = q(N) \mathbf{1}$ was then formed by 91 matrix–vector products in exact integer arithmetic and $\mathbf{v}^\top N^j w$ evaluated for $j = 0, 1, \dots$, again exactly. Every one of those integers vanishes from the index corresponding to $n = 91 + 1$ onwards, and the run continued until w was the zero vector, which settles all larger j at once. Lemma 3 gives the theorem. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the whole construction was compared against the entry’s DATA: the right-hand side of (4), divided by $4!$, was evaluated at every one of the 13 published indices and agrees exactly. This is a strong test of the modelling and not only of the arithmetic — the inversion of Lemma 1, the reading of the local condition, and the alignment of the entry’s index with the number of rows would each show up as a mismatch, and the quotient by $4!$ came out an integer at every index.

Second, the lumped chain was checked against the unlumped one: for the small shapes the walk counts $\mathbf{s}_i^\top M_i^m \mathbf{1}$ were computed directly over the full alphabet as well and agree with $\sum_P \varepsilon_P f_m(P)$ term by term.

Third, the recurrence (1) was evaluated on the published terms in exact integer arithmetic with no matrices involved, and the annihilation test of Lemma 3 was run until the working vector was identically zero rather than to a sampled prefix.

References

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